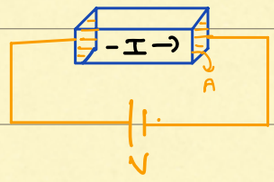
Anjan ; ESB 128

Quiz 1 (20%) Quiz 2 (20%) Endsem (50%)

Moodle Quiz (10%)

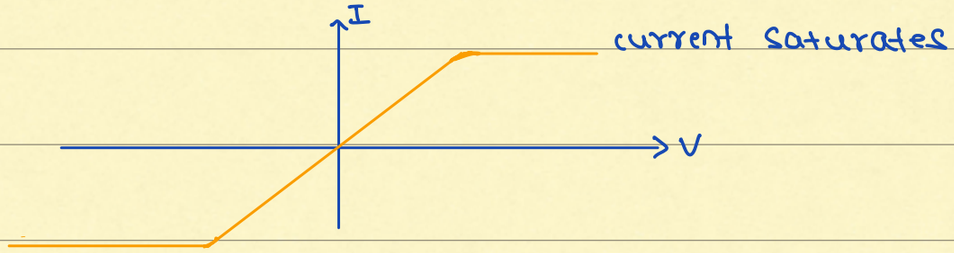
Books:- Semiconductor devices : Modelling & Technology

20101



$R = \frac{\rho L}{A}$

it is expected to have a linear I-V graph, but that's not the case

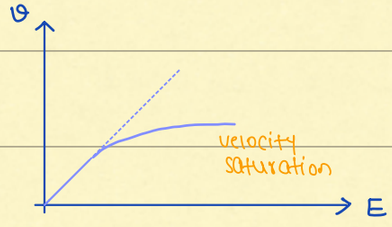


$n \rightarrow e^{\ominus}$ density $(1/cm^3)$

$\Rightarrow nq \rightarrow$ charge density (C/cm^3)

$\Rightarrow nqv \rightarrow$ current density $(A/cm^2) \Rightarrow \boxed{J = nqv}$

E (Electric field) = V/L



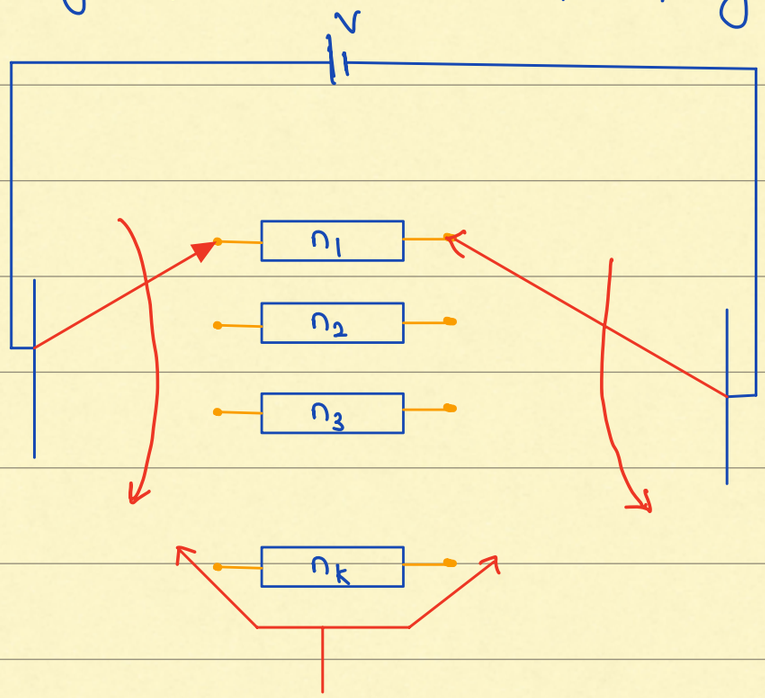
$v = \mu E$ $\quad \mu \rightarrow cm^2/V.sec$

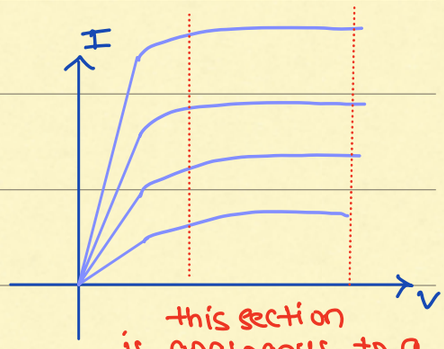
$J = nq\mu E$ (μ reduces as E increases inorder for velocity)

we know $J = \sigma E$

$\sigma \rightarrow$ conductivity $\Rightarrow \boxed{\sigma = nq\mu}$ $\quad n \rightarrow [10^5, 10^{19}]$

'n' can be altered / controlled by Temperature, light, Electric Field, Doping





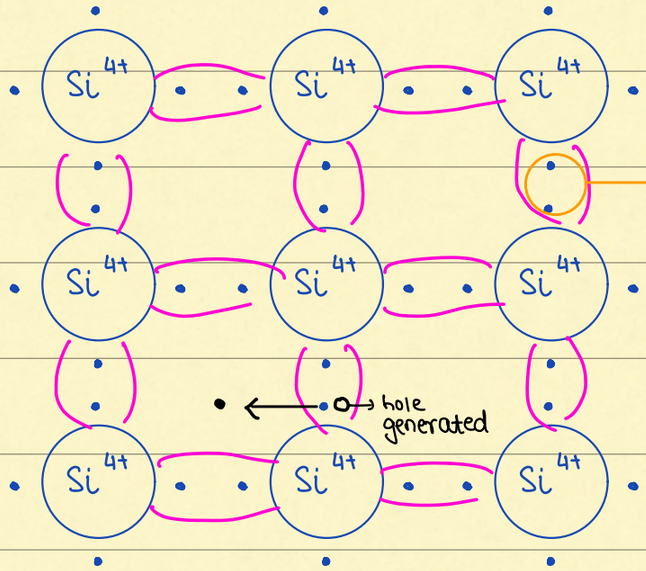
this section is analogous to a current source

Bond Model

Si: $1s^2 2s^2 2p^6 3s^2 3p^2$ 4VE

Ge: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^2$ 4VE

Atomic density of Si: 10^{22} atoms/cc



(intrinsic semiconductor)

potential ⇒ bonds have carriers to be broken

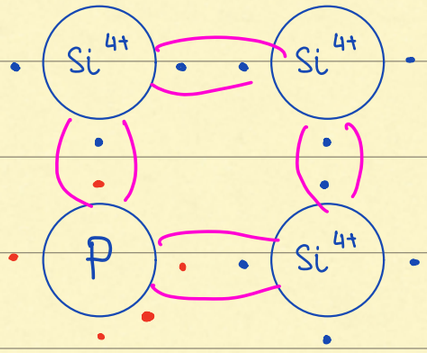
→ Gaps in bond can be part of current flow

→ Electrons in bond can't be part of current flow

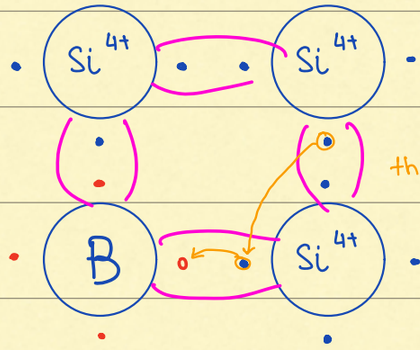
At T = 300K

$n = p \cong 10^{10}$/cc

21/01



replace 10^{16} Si atoms/cc with 10^{16} P atoms/cc (doping)



these gaps are freely moving

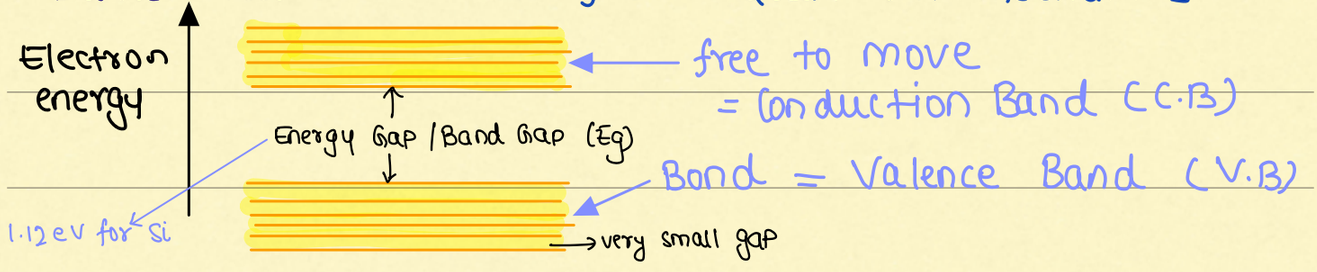
$\sigma = pq\mu$

If pentavalent atom is used to dope ⇒ we get more free e^-s. If trivalent atom is used to dope → we get more gaps.

Pentavalent ⇒ n-type semiconductors

Trivalent ⇒ p-type semiconductors

Band model: Borrow from Quantum mechanics

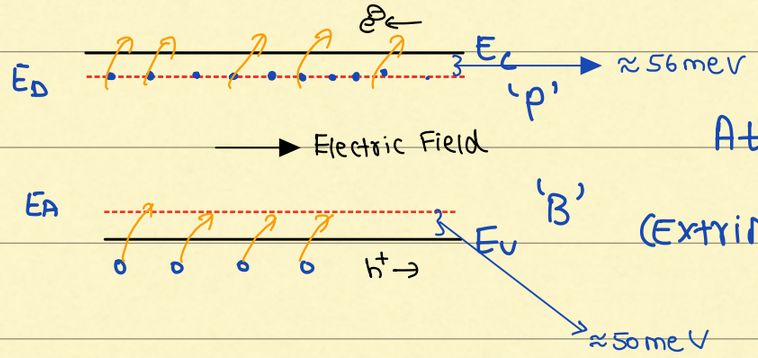


$1\ eV = 1.6 \times 10^{-19}\ J$

$E_g = 1.12\ eV$ for Si

$E_g = 0.68\ eV$ for Ge

$E_g = 1.43\ eV$ for GaAS



At 300K; $KT \cong 26$meV

(Extrinsic Semiconductor)

Generation process (G) Recombination process (R)

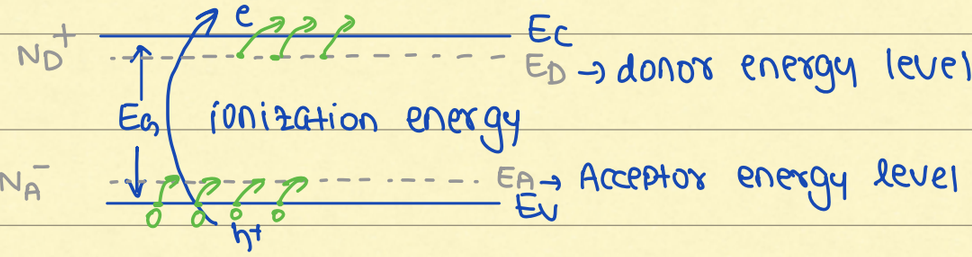
$G = R$ ← Equilibrium (Thermal Equilibrium)

Shallow dopant; Deep dopant → decides $E_c - E_D$ / $E_A - E_v$

$G \propto$ Temperature (T) ; $R \propto np$

23101

n_i is more if Temperature is more and E_g is less



$n_i = 10^{10}/cm^3$ @ $T = 300K$

↑ Intrinsic carrier concentration

$n_i = p = n$ for intrinsic semiconductors

$N_D = 10^{16}/cm^3 \Rightarrow$ complete ionization (all N_D become N_D^+)

$\Rightarrow n = 10^{10} + 10^{16} = 10^{16}$

$p = 10^{10}$

$R \propto np \propto 10^{26}$ But $G = 10^{20}$; so R has to reduce for equilibrium, i.e reduce $e^\ominus$s or $h^\oplus$s by recombination

1. $e^\ominus$s reduce $\Rightarrow p = 10^{10} \Rightarrow n = 10^{10} \Rightarrow 10^{16} - 10^{10} = 10^{16}$ $e^\ominus$s have to recombine with $h^\oplus$s but there are not 10^{16} $h^\oplus$s.
2. $h^\oplus$s reduce $\Rightarrow n = 10^{16} \Rightarrow p = 10^4 \Rightarrow 10^{10} - 10^4 = 10^{10}$ $h^\oplus$s have to recombine and there are 10^{10} $e^\ominus$s to recombine

$\Rightarrow n = 10^{16}/cm^3$; $p = 10^4/cm^3$ $\quad np = 10^{20}$ → Mass Action law

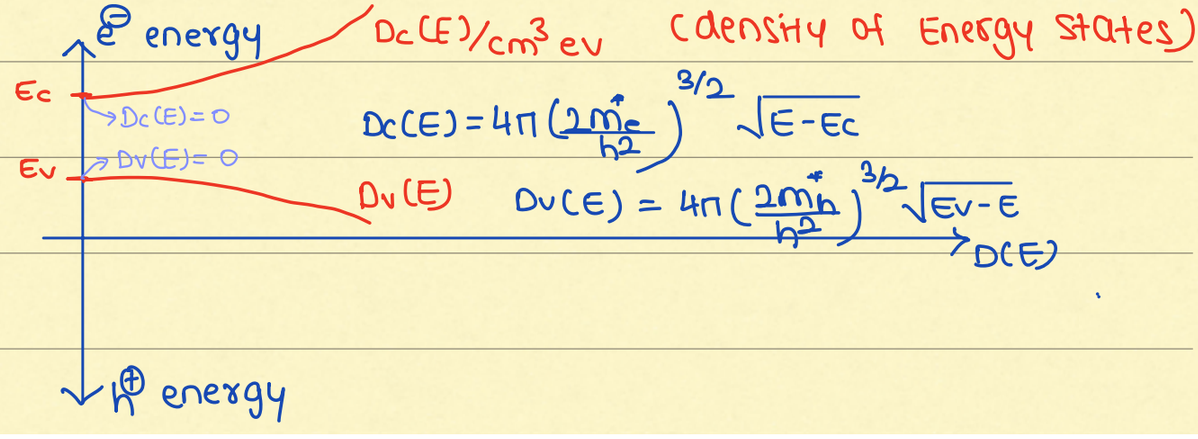
By charge conservation: $\boxed{p + N_D^+ = n + N_A^-}$ → Charge Neutrality law

$N_A = 10^{18}$; $N_D = 10^{16} \Rightarrow n + 10^{16} = p + 10^{18}$; $np = 10^{20}$

$\Rightarrow n = 10^{18}$; $p = 10^2$

$$\boxed{\sigma = nq\mu_n + pq\mu_p}$$

$\mu_n \rightarrow e^\ominus$ mobility ; $\mu_p \rightarrow h^\oplus$ mobility

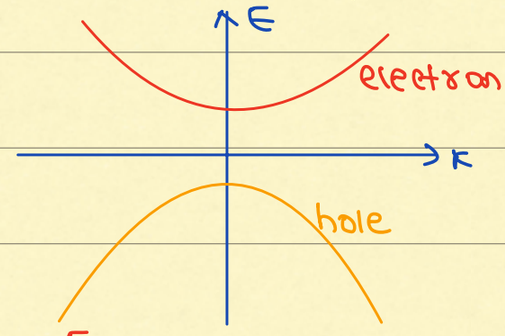


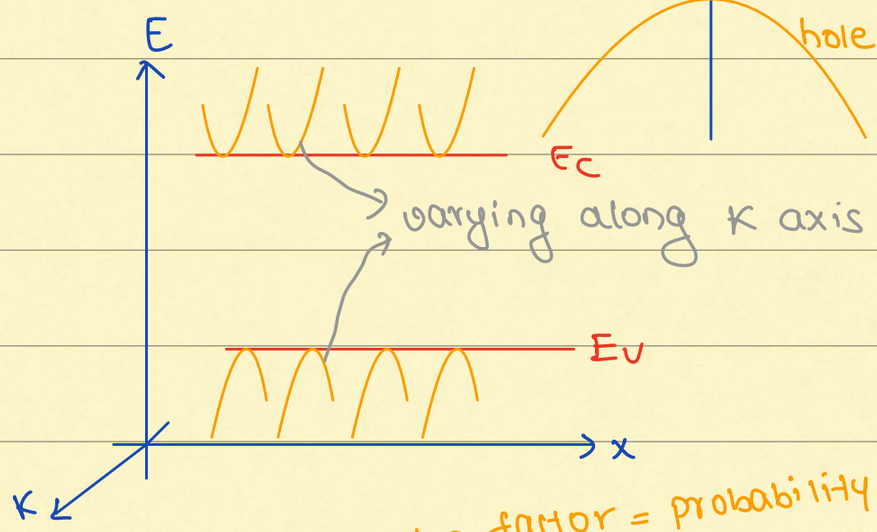
27/01

$p = \frac{h}{\lambda} \times \frac{2\pi}{2\pi} = \hbar k \Rightarrow$ KE of free particle $= \frac{\hbar^2 k^2}{2m}$

$\frac{d^2E}{dk^2} = \frac{\hbar^2}{m} \Rightarrow$ $\boxed{m = \hbar^2 \left(\frac{\partial^2 E}{\partial k^2}\right)^{-1}}$

more curvature, less m





- $n = \int_{E_c}^{E_c \text{ top}} D_c(E)\, dE\, f(E)$ → occupation factor = probability of occupation of an energy level by $e^\ominus$

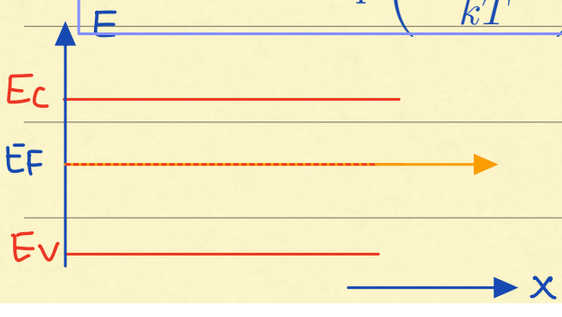
$D_c(E)$: no of energy levels in energy dE

- $p = \int_{(E_V)_{bottom}}^{E_V} D_V(E)\, dE\, (1 - f(E))$

$f(E)$ → fermi-dirac distribution function

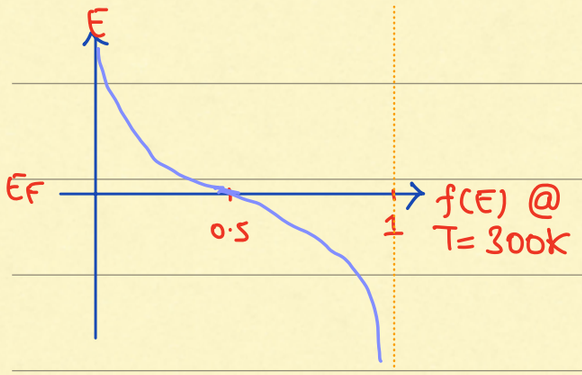
$$\boxed{f(E) = \frac{1}{1 + \exp\left(\frac{E - E_F}{kT}\right)}}$$

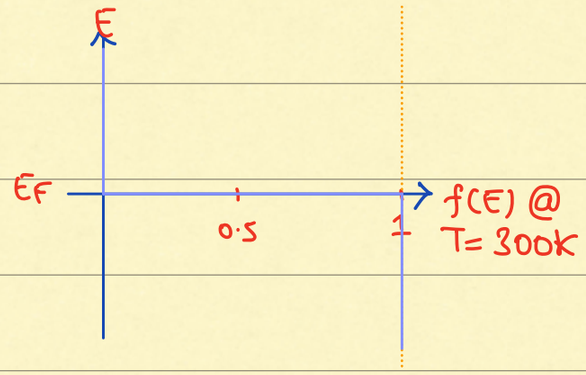
E_F → fermi energy level (measure of average energy of $e^\ominus$ in a system)



if $T \to 0$; $E \approx E_F^+ \Rightarrow f(E) = 0$

if $T \to 0$; $E \approx E_F^- \Rightarrow f(E) = 1$

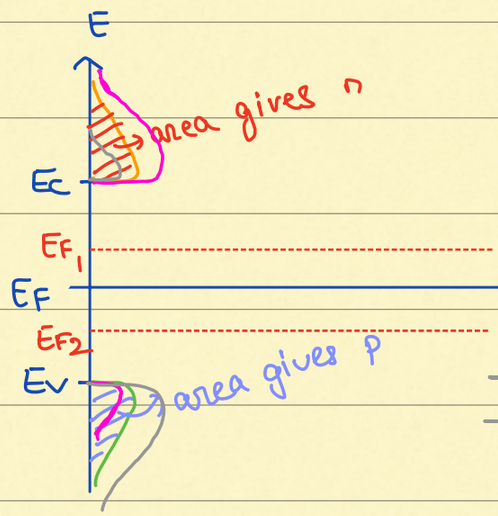




if $\exp\left(\frac{E-E_F}{KT}\right)$ is sufficiently large → $f(E) = \exp\left(-\frac{E-E_F}{KT}\right)$

$$E = E_f + \ln\left(\frac{1}{f(E)} - 1\right)$$

28/01



$D(E) \cdot f(E)$

$D(E)(1-f(E))$

- → n-type (n >> p)
- → new n, p after E_{F_1} change
- → p-type (p >> n)
- → new n, p after E_{F_2} change

If we change E_F; f(E) graph changes but not D(E) (D.O.S doesn't depend on f(E))

$$n = \int_{E_c}^{E_c \text{ top}} D_c(E) f(E)\, dE \qquad p = \int_{E_{v\,\text{bottom}}}^{E_v} (1-f(E))\, D_v(E)\, dE$$

$$D_c(E) = 4\pi\left(\frac{2m_e^*}{h^2}\right)^{3/2}\sqrt{E-E_c} \qquad D_v(E) = 4\pi\left(\frac{2m_h^*}{h^2}\right)^{3/2}\sqrt{E_v-E}$$

m_e^*, m_h^* → D.O.S effective mass ($m_e^* > m_h^*$)

m_n^*, m_p^* → Mobility effective mass ($m_n^* < m_p^*$)

If $E - E_F \geq 3KT$; we make maxwell boltzman approximation near conduction band ; and if $E_F - E \geq 3KT$ we can do the same near valence band.

In conduction band: $f(E) \approx \exp\left(-\frac{E-E_F}{KT}\right)$

$$n = \int_{E_c}^{E_{c\,top}} \frac{1}{1+\exp\left(\frac{E-E_F}{KT}\right)} \cdot 4\pi \left(\frac{2m_e^*}{h^2}\right)^{3/2} \sqrt{E-E_c}\, dE$$

$$= \int_{E_c}^{E_{c\,top}} \exp\left(-\frac{E-E_F}{KT}\right) 4\pi \left(\frac{2m_e^*}{h^2}\right)^{3/2} \sqrt{E-E_c}\, dE$$

$$= 4\pi \left(\frac{2m_e^*}{h^2}\right)^{3/2} \exp\left(\frac{-E_c+E_F}{KT}\right) \int_{E_C}^{E_{c\,top}} \exp\left(\frac{E_c-E}{KT}\right) \sqrt{E-E_c}\, dE$$

$\frac{E-E_c}{KT} = u$ substitution

$$= 4\pi \left(\frac{2m_e^*}{h^2}\right)^{3/2} \exp\left(\frac{E_F-E_c}{KT}\right) (KT)^{3/2} \int_0^{\frac{E_{c\,top}-E_c}{KT}} \sqrt{u} \exp(u)\, du$$

assume $E_{c\,top} = \infty$

$$= 4\pi \left(\frac{2m_e^* KT}{h^2}\right)^{3/2} \exp\left(\frac{E_F-E_c}{KT}\right) \int_0^{\infty} \sqrt{u}\, e^{-u}\, du$$

$\Gamma(3/2) = \frac{1}{2} \times \sqrt{\pi}$

$$n = 2\underbrace{\left(\frac{2\pi m_e^* kT}{h^2}\right)^{3/2}}_{N_C} \exp\left(\frac{E_F - E_c}{kT}\right) \qquad n = N_c \exp\left(\frac{E_F - E_C}{kT}\right)$$

In valence band: $1 - f(E) = \dfrac{\exp\left(\frac{E-E_f}{KT}\right)}{1 + \exp\left(\frac{E-E_F}{KT}\right)}$

If $E - E_F \geq 3KT \rightarrow 1 - f(E) = \exp\left(\frac{E-E_F}{KT}\right)$

$$p = \int_{E_v \text{ bottom}}^{E_v} \exp\left(\frac{E-E_F}{KT}\right) 4\pi \left(\frac{2m_h^*}{h^2}\right)^{3/2} \sqrt{E_v - E}\, dE$$

$$= 4\pi \left(\frac{2m_h^*}{h^2}\right)^{3/2} \exp\left(\frac{E_v - E_F}{KT}\right) \int_{E_v \text{ bottom}}^{E_v} \exp\left(\frac{E-E_v}{KT}\right) \sqrt{E_v - E}\, dE$$

$\frac{E_v - E}{KT} = u$ substitution

$$= 4\pi \left(\frac{2m_h^*}{h^2}\right)^{3/2} \exp\left(\frac{E_v - E_F}{KT}\right) \int_{\infty}^{0} \exp(-u) \sqrt{u} \left(\sqrt{KT}\right)^3 (-du)$$

$$= 4\pi \left(\frac{2KTm_h^*}{h^2}\right)^{3/2} \exp\left(\frac{E_v - E_F}{KT}\right) \int_0^{\infty} e^{-u} \sqrt{u}\, du$$

$\hookrightarrow \sqrt{\pi}/2$

$$p = 2\underbrace{\left(\frac{2\pi kTm_h^*}{h^2}\right)^{\frac{3}{2}}}_{N_V} \exp\left(\frac{E_v - E_F}{kT}\right) \qquad p = N_v \exp\left(\frac{E_v - E_F}{kT}\right)$$

30/01

$n = N_c f(E_c)$ — $N_c \rightarrow$ effective D.O.S @ C.B

$p = N_v(1 - f(E_v))$ — $N_v \rightarrow$ effective D.O.S @ V.B

at a given T ; $N_c > N_v$

At T = 300K ; $N_c \cong 3.2 \times 10^{19}/cm^3$; $N_v \cong 1.8 \times 10^{19}/cm^3$

for intrinsic semiconductor

$$n = n_i = N_c \exp\left(\frac{E_{fi} - E_c}{KT}\right) \Rightarrow \frac{N_c}{N_v} = \exp\left[\frac{E_v + E_c - 2E_{fi}}{KT}\right]$$

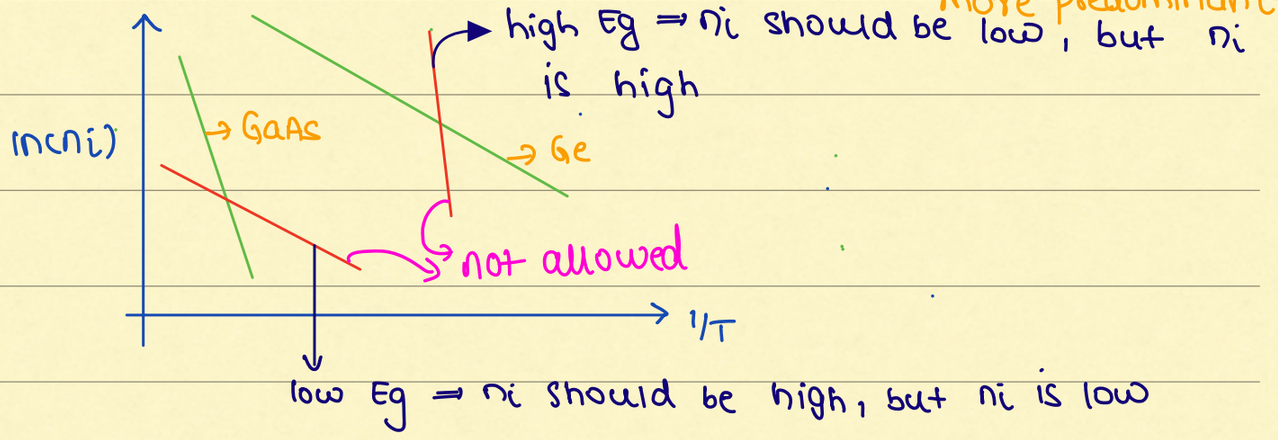
$$p = n_i = N_v \exp\left(\frac{E_v - E_{fi}}{KT}\right)$$

$$\Rightarrow E_{fi} = \frac{E_C + E_V}{2} - \frac{kT}{2}\ln\left(\frac{N_C}{N_v}\right) \qquad n_i = \sqrt{N_c N_v}\exp\left(-\frac{E_g}{2kT}\right)$$

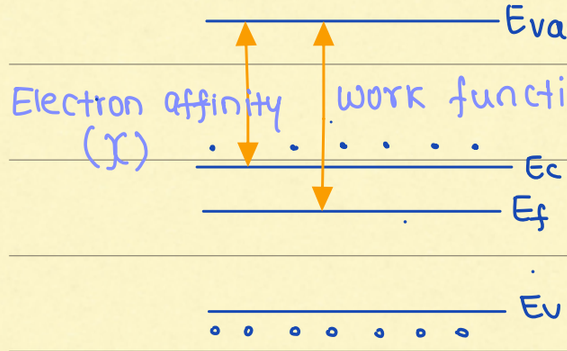
$$n = n_i \exp\left(\frac{E - E_{fi}}{kT}\right) \qquad p = n_i \exp\left(\frac{E_{fi} - E_f}{kT}\right) \qquad np = n_i^2$$

$$\ln(n_i) = \frac{\ln(N_c N_v)}{2} - \frac{E_g}{2KT} = c + a\ln(T) - \frac{b}{T}$$

↓ more predominant



Reality of Semiconductors:



$E_{vaccum} = q\psi_{electrostatic} = -q\psi$

$E_f = q\psi_{electrochemical} = -q\psi_F$

$\psi \rightarrow$ potential

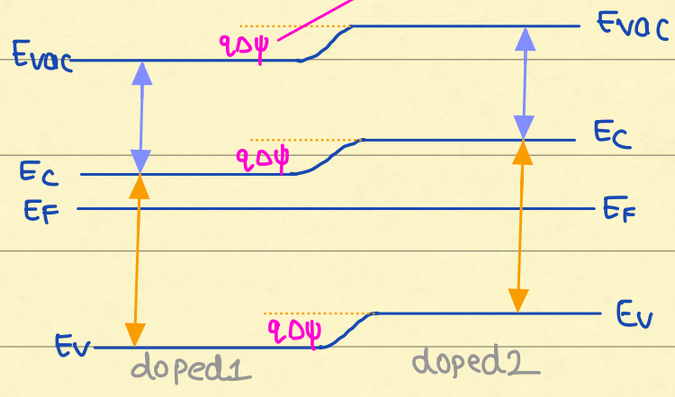
Existance of fermi level itself says that E_F is flat spacially. $\frac{d}{dx} E_F = 0$

Existance of fermi level itself says that the system is in equilibrium.

If E_F is flat current will not flow.

02/02

Built-in potential / Contact potential



(Thermal Equilibrium)

e's don't flow due to flat fermi level

Eg and qχ are same for both cases

Electrostatic field / Electric Field (ε): $\varepsilon = -\frac{d\psi}{dx}$ $E = -q\psi$

Charge density (ρ_{ch}): $\rho_{ch} = q[N_D^+ + p - n - N_A^-]$

$$\int_S \vec{\mathcal{E}}\cdot d\vec{s} = \frac{Q}{\epsilon} \Rightarrow \int_V \vec{\nabla}\cdot\vec{\mathcal{E}}\, dV = \frac{1}{\epsilon}\int_V \rho_{ch}\cdot dV \Rightarrow \vec{\nabla}\cdot\vec{\mathcal{E}} = \frac{\rho_{ch}}{\epsilon_{Si}}$$

$\epsilon_{Si} = 10^{-12}$ F/cm

If one-dimension: $\frac{d\mathcal{E}}{dx} = \frac{\rho_{ch}}{\epsilon_{Si}} \Rightarrow$ $$\frac{d^2\psi}{dx^2} = -\frac{\rho_{ch}}{E_{Si}}$$ (1)

poisson's equation

$np = n_i^2 \Rightarrow n(n + N_A^- - N_D^+) = n_i^2 \Rightarrow n^2 - n(N_D^+ - N_A^-) - n_i^2 = 0$

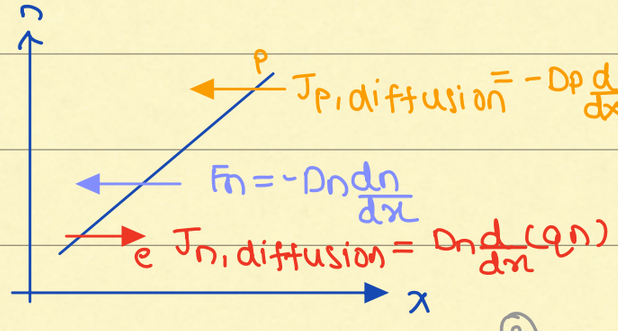
$$\Rightarrow n = \frac{N_D^+ - N_A^-}{2}\left(1 + \sqrt{1 + \left(\frac{2n_i}{N_D^+ - N_A^-}\right)^2}\right)$$

$$p = \frac{N_A^- - N_D^+}{2}\left(1 + \sqrt{1 + \left(\frac{2n_i}{N_A^- - N_D^+}\right)^2}\right)$$

03/05

Electron current density: $J_n = \sigma_n \varepsilon = nqv_{drift}$

Drift current $= (nq\mu_n + pq\mu_p)\varepsilon = (\sigma_n + \sigma_p)\varepsilon = \sigma\varepsilon$



	+	−
mobile	p	n
immobile	N_D^+	N_A^-

$$J_n = nq\mu_n\varepsilon + qD_n\frac{dn}{dx}$$ (2)

$$J_p = pq\mu_p\varepsilon - qD_p\frac{dp}{dx}$$ (3)

drift diffusion equation

D_n, D_p → diffusion constants / diffusivity (unit: v)

depends on temperature $\frac{D_n}{\mu_n} = \frac{kT}{q}$

$$D_n = \mu_n V_T$$

$$\frac{\partial}{\partial t}\rho_{ch} = -\vec{\nabla}\cdot\vec{J} \Rightarrow -\frac{\partial}{\partial t}(qn) = -\frac{\partial}{\partial x}J_n$$

$$\frac{d}{dt}(n) = \frac{1}{q}\frac{\partial}{\partial x}(J_n) + G_n - R_n \quad (4)$$

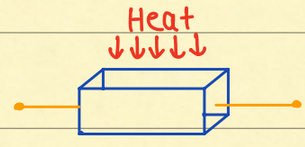
$$\frac{d}{dt}(p) = -\frac{1}{q}\frac{\partial}{\partial x}(J_p) + G_p - R_p \quad (5)$$

continuity equations

The whole semiconductor theory is the 5 equations and everything is an application of these.

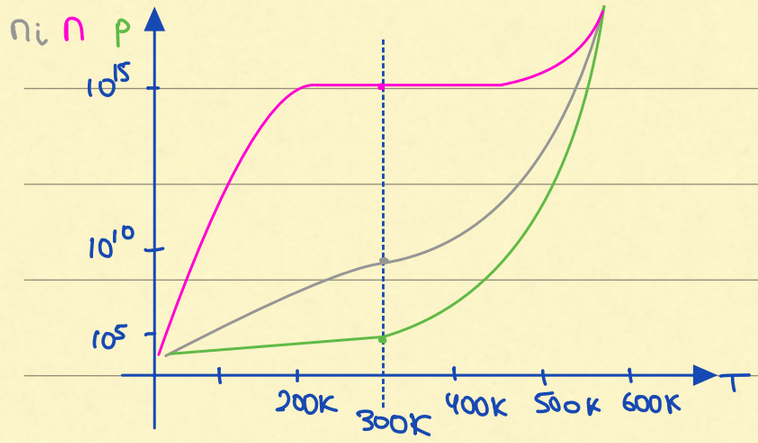
04/02

- $\frac{d^2\psi}{dx^2} = -\frac{\rho_{ch}}{E_{Si}}$
- $J_n = nq\mu_n\varepsilon + qD_n\frac{dn}{dx}$
- $J_p = pq\mu_p\varepsilon - qD_p\frac{dp}{dx}$
- $\frac{d}{dt}(n) = \frac{1}{q}\frac{\partial}{\partial x}(J_n) + G_n - R_n$
- $\frac{d}{dt}(p) = -\frac{1}{q}\frac{\partial}{\partial x}(J_n) + G_p - R_p$



Case I: n-type Si

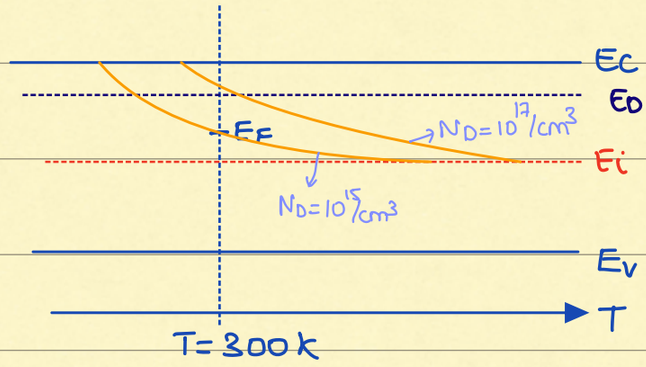
$N_D = 10^{15}/cm^3 = N_D^+ = n$. @ $T = 300K$; $n_i = 10^{10}/cm^3 \Rightarrow p = 10^5/cm^3$



$\frac{p}{n}$ increases as T↑

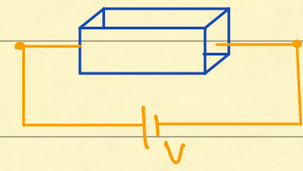
eventually $p/n \to 1$

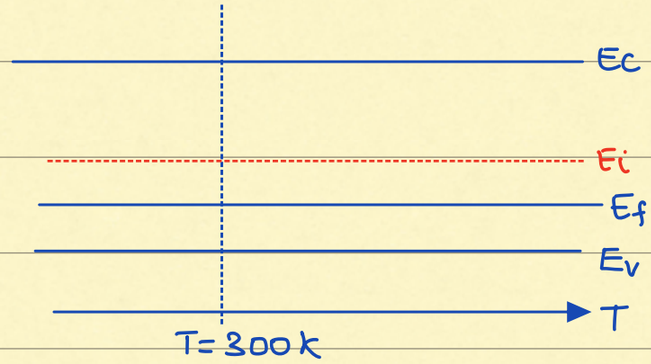
(intrinsic)



Case II: p-type

$N_A = N_A^- = 10^{15}/cm^3 \Rightarrow n = 10^5/cm^3$





Quasi fermi level (IMREF)

· for $e^\ominus$ - E_{fn} · for $h^\oplus$ - E_{fp}

↓

under non equilibrium

$$n = n_i \exp\left(\frac{E_{fn} - E_i}{kT}\right)$$

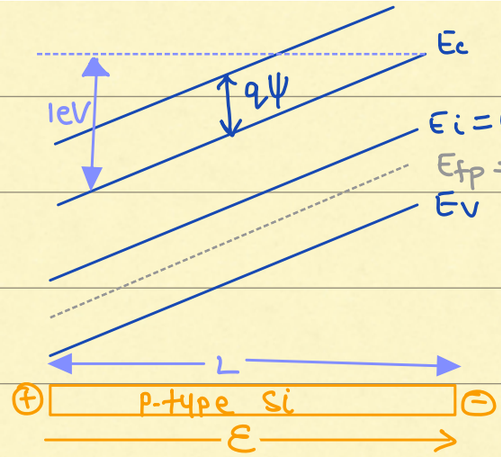
$$p = n_i \exp\left(\frac{E_i - E_{fp}}{kT}\right)$$

$$np = n_i^2 \exp\left(\frac{E_{fn} - E_{fp}}{kT}\right)$$

If $E_{fn} = E_{fp} \Rightarrow np = n_i^2$

If $E_{fn} > E_{fp} \Rightarrow np > n_i^2 \Rightarrow R > G$

If $E_{fn} < E_{fp} \Rightarrow np < n_i^2 \Rightarrow R < G$



"e's can roll"

$V = 1$ volt $\Rightarrow \psi \rightarrow$ straight line

$$\frac{d^2\psi}{dx^2} = 0 \qquad E_{vac} = q\psi$$

$$\mathcal{E} = -\frac{d\psi}{dx} = \frac{V}{L}$$

09/02

$$J_p = pq\mu_p\mathcal{E} - qD_p\frac{dp}{dx} = pq\mu_p\mathcal{E}$$

$$p = n_i \exp\left(\frac{E_i - E_{fp}}{KT}\right) \qquad \frac{dp}{dx} = \frac{p}{KT}\left[\frac{d}{dx}(E_i - E_{fp})\right]$$

$$\mathcal{E} = -\frac{d\psi}{dx} = -\frac{d}{dx}\left[-\frac{1}{q}E_{vac}\right] = \frac{1}{q}\frac{d}{dx}E_{vac} = \frac{1}{q}\frac{d}{dx}E_i$$

same slope

$$J_p = pq\mu_p\left(\frac{1}{q}\frac{dE_i}{dx}\right) - q\left(\mu_p\cdot\frac{KT}{q}\right)\left(\frac{p}{KT}\frac{d}{dx}(E_i - E_{fp})\right)$$

$$\boxed{J_p = p\mu_p\frac{d(E_{fp})}{dx}}$$

Drift diffusion equation compact form

drift diffusion transfer model for hole

$$J_n = nq\mu_n\mathcal{E} + qD_n\frac{dn}{dx}$$

$$n = n_i \exp\left(\frac{E_{fn} - E_i}{KT}\right) \qquad \frac{dn}{dx} = \frac{n}{KT}\frac{d}{dx}(E_{fn} - E_i)$$

$$\mathcal{E} = -\frac{d\psi}{dx} = \frac{1}{q}\frac{d(E_i)}{dx}$$

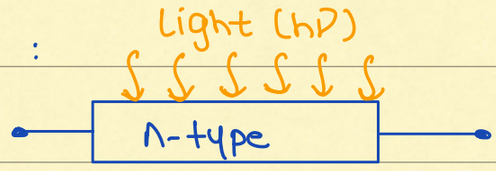
$$J_n = nq\mu_n\left(\frac{1}{q}\frac{dE_i}{dx}\right) + q\left(\mu_n\cdot\frac{KT}{q}\right)\left(\frac{n}{KT}\frac{d(E_{fn} - E_i)}{dx}\right)$$

$$\boxed{J_n = n\mu_n\frac{d(E_{fn})}{dx}}$$

Drift diffusion equation compact form

drift diffusion transfer model for $\ominus$

Case III :



$h\nu > E_G$ to generate $e^\ominus$ hole pairs

T = 300K (assume no light energy converts into heat)

$N_D = N_D^+ = n_{n_0} = 10^{16}/cm^3 \Rightarrow p_{n_0} = 10^4/cm^3$

$\Delta n_n = \Delta p_n = 10^{12}/cm^3$ ($e^\ominus$ - $h^\oplus$ pairs are generated)

$n = n_{n_0} + \Delta n_n = 10^{16}/cm^3 \qquad p = p_{n_0} + \Delta p_n = 10^{12}/cm^3$

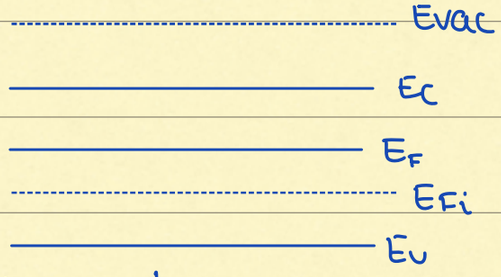
$np = 10^{28}/cm^6 > n_i^2$ (not equilibrium)

$= n_i^2 \exp\left(\frac{E_{fn} - E_{fp}}{KT}\right)$

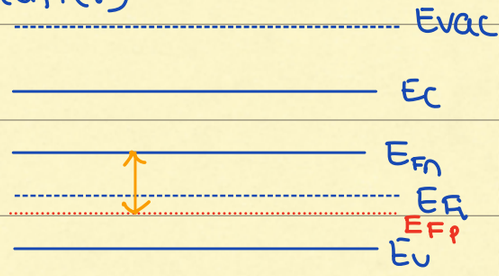
E_{Fn} → will remain @ same place

E_{Fp} → goes below E_{Fi}

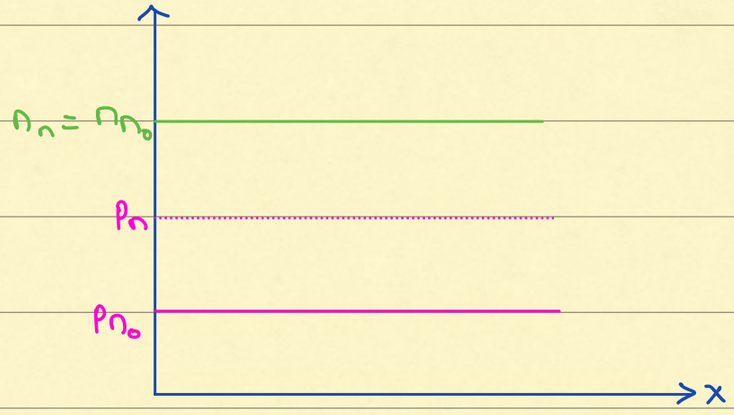
(before)



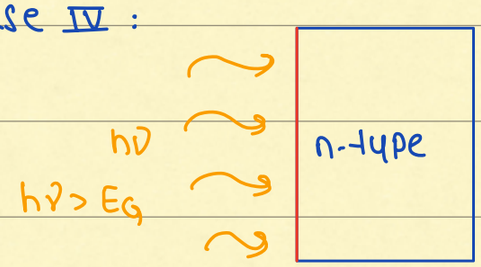
(after)



$\frac{d}{dx} E_{Fp} = \frac{d}{dx} E_{Fn} = 0 \Rightarrow J_n = J_p = 0$



Case IV :

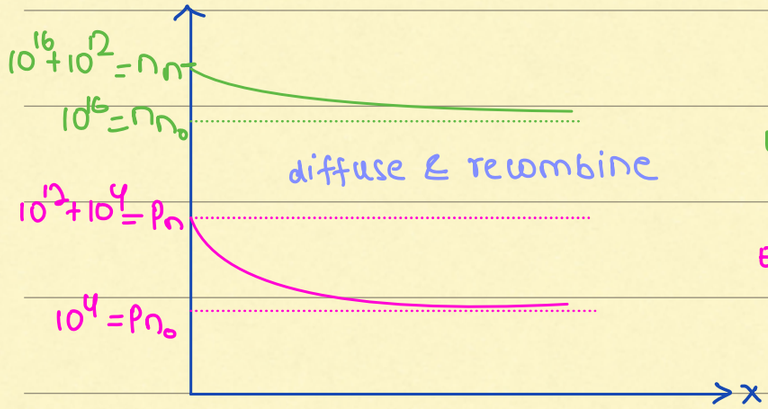


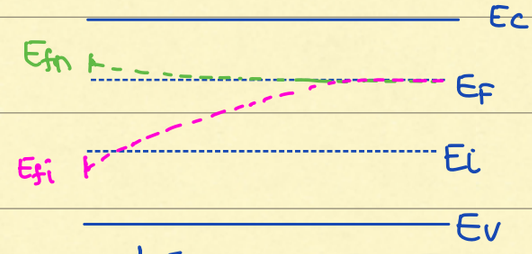
$n_{no} = 10^{16}/cm^3$

$p_{no} = 10^{4}/cm^3$

$\Delta n_n\big|_s = \Delta p_n\big|_s = 10^{12}/cm^3$

surface generation of $e^{\ominus}$-$h^{\oplus}$ pairs



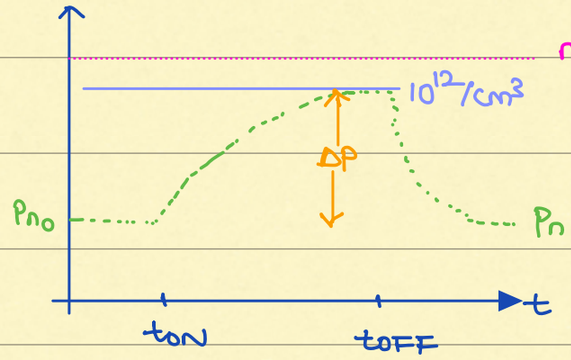


$\frac{d}{dx}E_F = 0 \Rightarrow J_n = J_p = 0$

$\frac{d}{dy}E_{Fn} < 0 \Rightarrow J_n < 0$

$\frac{d}{dx}E_{Fp} > 0 \Rightarrow J_p > 0$

10102



$$\frac{d}{dt}(p) = -\frac{1}{q}\frac{\partial}{\partial x}(J_p) + G_p - R_p + G_L$$

(0 ; thermal ; Given (light))

$J_p = 0$ $\quad R \propto n_n p \Rightarrow R = \beta n_n p_n$

$G_p = R_{p_{no}} = \beta n_{no} p_{no}$ (under thermal equilibrium)

$$\frac{d(p_n)}{dt} = \frac{d\,p_{n_0}}{dt}^{0} + \frac{d(\Delta p)}{dt}$$

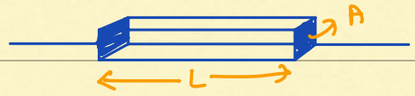
$$\Rightarrow \frac{d}{dt}(\Delta p_n) = \beta n_{n_0} p_{n_0} - \beta (n_{n_0} + \Delta p_n)(p_{n_0} + \Delta p_n) + G_L$$

$$\Rightarrow \frac{d}{dt}(\Delta p_n) = G_L - \beta (\Delta p_n \cdot p_{n_0} + \Delta p_n \cdot n_{n_0} + \Delta p_n^2)$$

$$= G_L - \beta n_{n_0} \Delta p_n \qquad \Delta p_n = \Delta n_n << n_{n_0}$$

$$= G_L - \frac{\Delta p_n}{\tau_p} \qquad \beta \to cm^3/s \quad \beta n_{n_0} \to 1/s$$

$\tau_p \to$ recombination lifetime (avg lifetime before recombination) ~ 1 μs (slow)



$R = \frac{\rho L}{A} \quad C = \frac{\varepsilon_{Si} A}{L} \Rightarrow RC = \varepsilon_{Si} \times \rho \sim 1$ ps (fast)

relaxation time constant

$$\frac{d(\Delta p_n)}{dt} = G_L - \frac{\Delta p_n}{\tau_p} \Rightarrow \frac{d(\Delta p_n)}{(-G_L \tau_p + \Delta p_n)} = -\frac{dt}{\tau_p}$$

$$\Rightarrow \ln\left(\frac{\Delta p_n - G_L \tau_p}{\Delta p_{n_0}^{0} - G_L \tau_p}\right) = -\frac{t}{\tau_p}$$

$$\Rightarrow \Delta p_n - G_L \tau_p = (-G_L \tau_p)\, e^{-t/\tau_p}$$

$$\Rightarrow \boxed{\Delta P_n = G_L \tau_p \left(1 - e^{-t/\tau_p}\right)} \qquad t < t_{OFF}$$

$$\boxed{\Delta P_n = G_L \tau_p \left(e^{-t/\tau_p}\right)} \qquad t > t_{OFF}$$

In steady state:

$$\frac{d}{dt}(n) = \frac{1}{q}\frac{\partial}{\partial x}(J_n) + G_n - R_n = 0 \quad -①$$

$\frac{d}{dt}$ → 0 (no time dependence)

$$\frac{d}{dt}(p) = -\frac{1}{q}\frac{\partial}{\partial x}(J_p) + G_p - R_p = 0 \quad -②$$

Subtract ① and ② $\Rightarrow \frac{d}{dx}(J_n + J_p) = 0 \Rightarrow J_n + J_p = J(x)$ → constant = 0

$J_p(x) = -J_n(x)$

$\Rightarrow q p \mu_p \mathcal{E} - q D_p \frac{dp}{dx} = -q n \mu_n \mathcal{E} - q D_n \frac{dn}{dx}$

$\Rightarrow \mathcal{E}(q p \mu_p + q n \mu_n) = q D_p \frac{dp}{dx} - q D_n \frac{dn}{dx}$

Approximation: $\frac{dn_n}{dx} = \frac{dp_n}{dx}$ $(D_n > D_p)$

$$\mathcal{E} = \frac{q\frac{dp}{dx}(D_p - D_n)}{q(p\mu_p + n\mu_n)} = \frac{\frac{dp}{dx}(D_p - D_n)}{p\mu_p + n\mu_n}$$

$$\frac{J_{p,\text{drift}}}{J_{p,\text{diffusion}}} = \frac{pq\mu_p \mathcal{E}}{-qD_p\frac{dp}{dx}} = \frac{p\mu_p(D_n - D_p)}{D_p(n\mu_n + p\mu_p)} \ll 1$$

⇒ drift component of minority carrier can be ignored

11/02

case iv: steady state illumination from one side $\Rightarrow \frac{dp}{dt} = -\frac{1}{q}\frac{dJ_p}{dx} + G_p - R_p = 0$ $(G_L = 0$ inside$)$

$$-\frac{1}{q}\frac{d(-qD_p\frac{d\Delta p_n(x)}{dx})}{dx} + \left(-\frac{\Delta p_n(x)}{\tau_p}\right) = 0$$

$$\Rightarrow \frac{d^2(\Delta p_n(x))}{dx^2} = \frac{\Delta p_n(x)}{D_p \tau_p}$$

$$\frac{d^2(\Delta p_n(x))}{dx^2} = \frac{\Delta p_n(x)}{L_p^2}$$

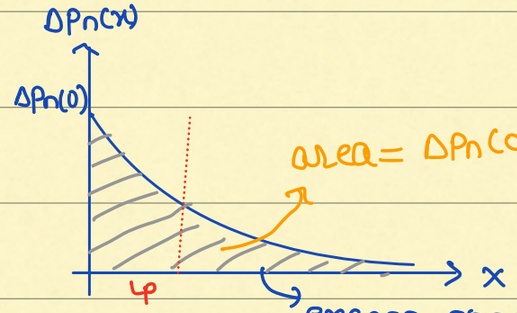
$L_p \rightarrow$ diffusion length $= D_p\tau_p$

$$\Rightarrow \Delta p_n(x) = Ae^{-x/L_p} + Be^{x/L_p}$$

$x \to \infty$; $\Delta p_n(x) = 0 \Rightarrow B = 0$

$x = 0$; $\Delta p_n(x) = \Delta p_{n_0} \Rightarrow \Delta p_n(x) = \Delta p_{n_0} e^{-x/L_p}$

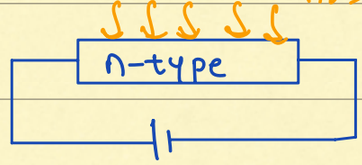
$$J_p(0) = -qD_p\frac{d(\Delta p_n(x))}{dx}\Big|_{x=0} = \frac{q \cdot D_l \cdot \Delta p_{n_0}}{L_p}$$



$L_p \rightarrow$ average length of diffusion before recombination

excess charge $= qA\Delta p_n(0).L_p$

case V :



$n_{n_0} = 10^{16}/cm^3$

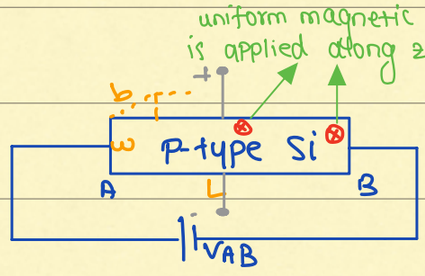
$p_{n_0} = 10^4/cm^3$

$$I = \frac{V}{R} \quad R = \frac{\rho L}{A} \quad \rho = \frac{1}{\sigma} = \frac{1}{q(n_n\mu_n + p_n\mu_p)}$$

↓ neglected

→ bad light sensor

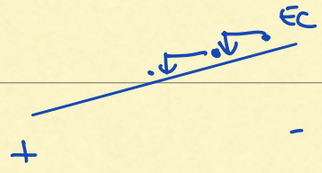
case VI :



B creates an e-field along y-axis and hence a potential difference

13102

Remember:



where $e^{\ominus}$s flow due to energy difference. We get dissipation of heat I^2R due to energy released by $e^{\ominus}$ in the lattice

Lattice scatter ⟶ depends on T

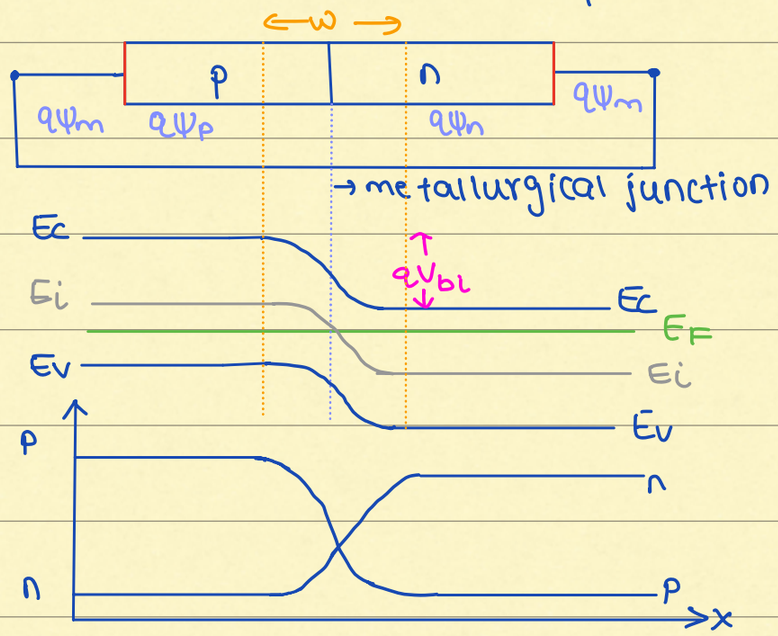
Impurity Scatter → depends on N

→ more @ lower T since V_{TH} is not as much at lower T so it can be affected more.

$N_A^- = N_D^+$ ⟹ doesn't mean intrinsic (different mobilities)

23102

W → depletion width



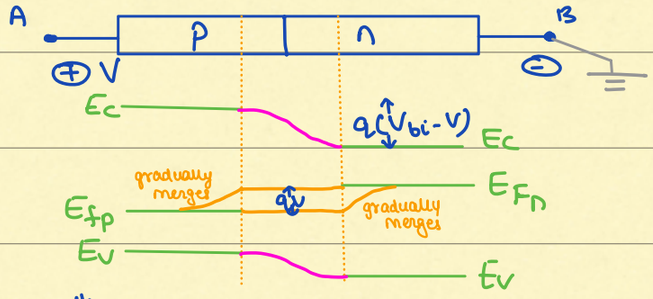
$V_{bi} = V_T \ln\left(\frac{N_A N_D}{n_i^2}\right)$ (assume all are ionized)

$T \uparrow \Rightarrow E_i$ comes close to $E_F \Rightarrow V_{bi}$ will reduce

$$W = \sqrt{\frac{2\epsilon_{Si} V_{bi} (N_A + N_D)}{qN_A N_D}} = \sqrt{\frac{2\epsilon_{Si} V_{bi}}{qN_{eff}}}$$

- If no contact plates: $\phi = \psi_p - \psi_n$
- But since we put contact plates we get $\phi = 0$ always (we can't measure using voltmeter)
- Voltmeter measures fermi-level difference

Biased P-n junctions

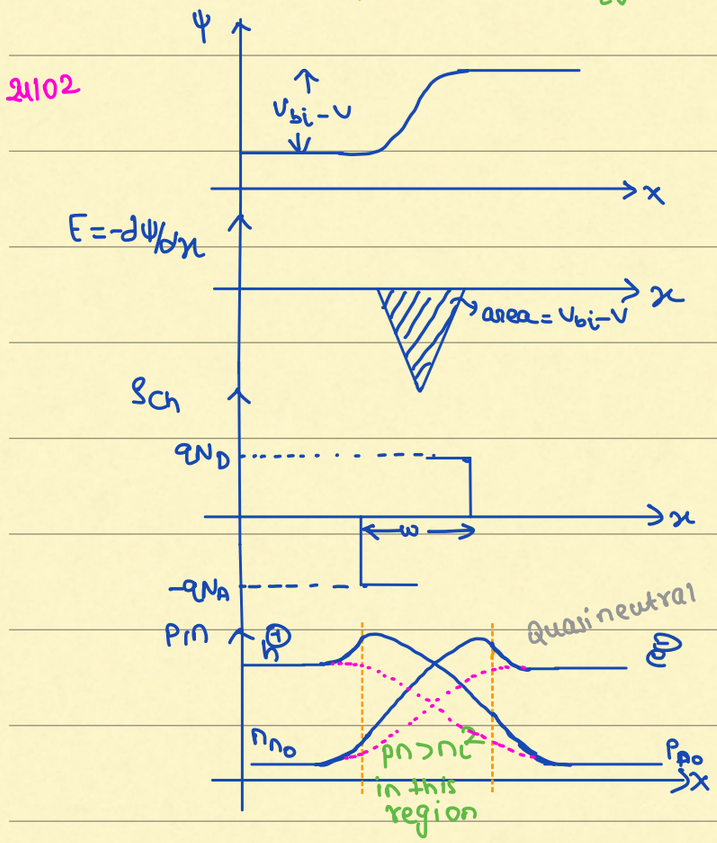


Forward bias

$$J_p = p\mu_p \frac{d(E_{fp})}{dx}$$

$$J_n = n\mu_n \frac{d(E_{fn})}{dx}$$

2/10/2



$$\frac{dE}{dx} = \frac{\rho_{ch}}{\epsilon_{si}}$$

$$W = \sqrt{\frac{2\epsilon_{si}(V_{bi}-V)}{qN_{eff}}}$$

low level injection:

$$n_n(x_n) \approx n_{no}$$

$$p_p(-x_p) \approx p_{po}$$

$$V_{bi} = V_T \ln\left(\frac{p_{po}}{p_{no}}\right)$$

$$p_{no} = p_{po} \exp\left(\frac{-V_{bi}}{V_T}\right)$$

$$p_n(x_{no}) = p_{no} = p_p(-x_{po})\, e^{-V_{bi}/V_T}$$

$$p_n(x_n) = p_{no} + \Delta p_n(x_n) = p_p(-x_{po})\, e^{-(V_{bi}-V)/V_T}$$

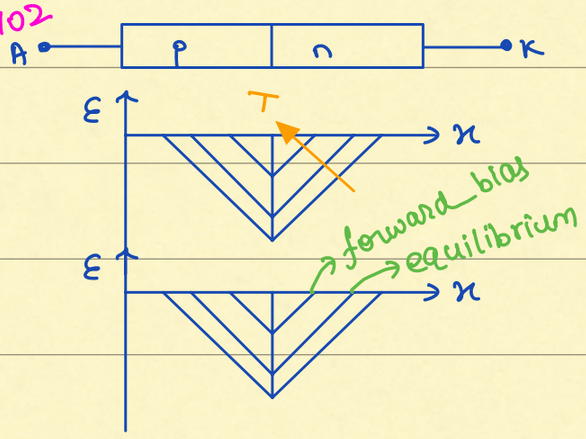
$$p_n(x_n) = p_{no} \exp\left(-\frac{V_{bi}-V}{V_T}\right)$$

$$\frac{p_{no} + \Delta p_n(x_n)}{p_{no}} = \exp\left(\frac{V}{V_T}\right)$$

$$\Delta p_n(x_n) = p_{no}\left(e^{V/V_T} - 1\right)$$

$$\Delta n_p(-x_p) = n_{po}\left(e^{V/V_T} - 1\right)$$

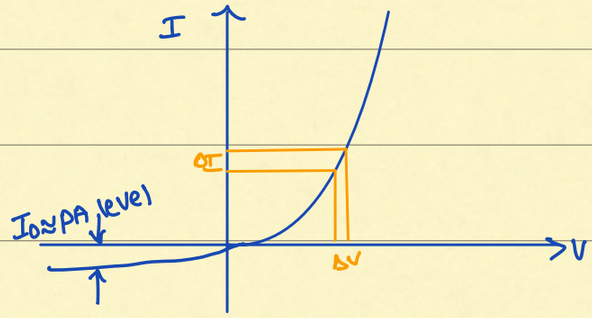
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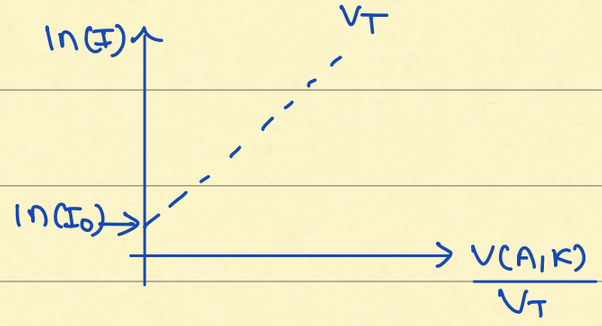
27/02



$$I = I_0\left[e^{V(A,K)/V_T} - 1\right]$$

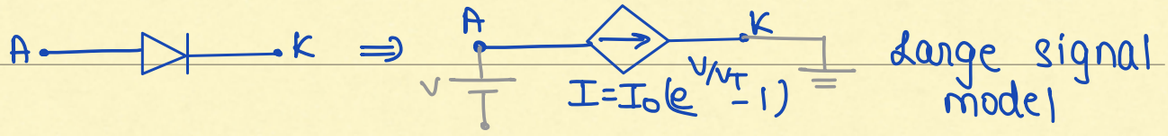


$$\ln I \approx \ln I_0 + \frac{V(A,K)}{V_T}$$

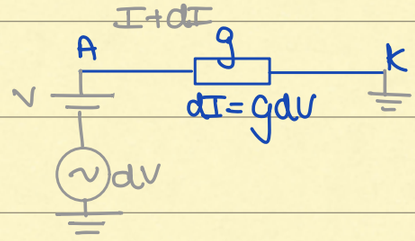


$$\frac{dI}{dV} = g = \frac{I + I_0}{V_T} \approx \frac{I}{V_T}$$ (increases with current)

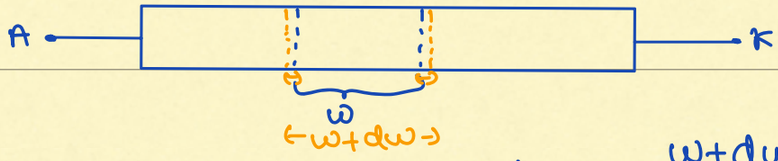
↑ conductance of diode



large signal model



small signal model



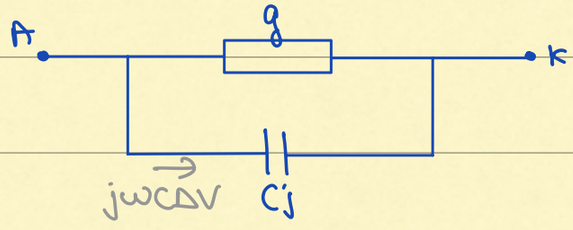
like a capacitor

$$C_j = \frac{\epsilon_{Si} A}{W}$$

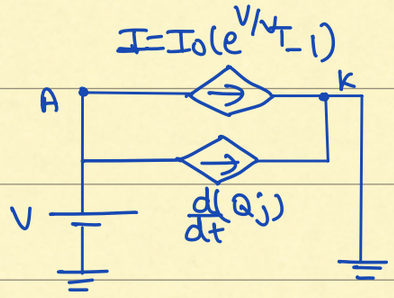
$$W = \sqrt{\frac{2\epsilon_{Si}(V_{bi} - V)}{qN_{eff}}}$$

$$W + dW = \sqrt{\frac{2\epsilon_{Si}(V_{bi} - V - dV)}{qN_{eff}}}$$

$C_{j0} = \frac{\epsilon A}{W_0}$ ↑ under no bias V=0 $\Rightarrow \frac{C_j}{C_{j0}} = \frac{1}{\sqrt{1 - V/V_{bi}}}$ V → -ve (reverse)



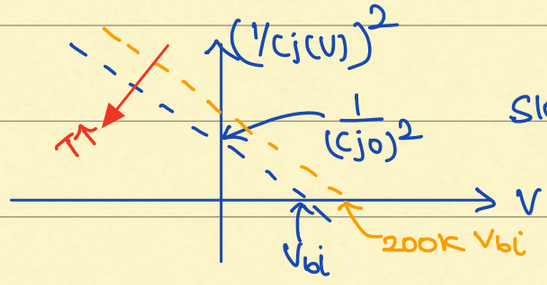
small-signal model



$$Q_J = \int_0^V C_j(V)\,dV$$

large signal model

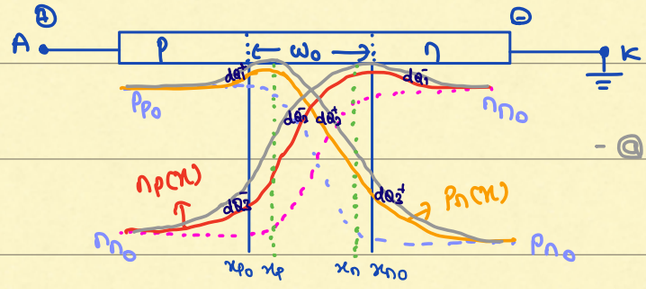
$$C_j(V) = \frac{C_j(0)}{\sqrt{1-V/V_{bi}}} = \frac{1}{\{C_j(V)\}^2} = \frac{1}{\{C_{j0}\}^2}\left\{1-\frac{V}{V_0}\right\}$$



$$\text{Slope} = -\frac{1}{(C_{j0})^2 V_{bi}} = \frac{-\omega_0^2}{\epsilon_{Si}^2 A^2}\frac{1}{V_{bi}}$$

= independent of V_{bi}

2/3/26



- @ V+dV

$dQ_1^+ = dQ_3^-$

$dQ_3^+ = dQ_2^-$

$dQ_2^+ = dQ_1^-$

$$C_{diode} = \frac{dQ_1^+ + dQ_2^+ + dQ_3^+}{dV} = \frac{dQ_1^+ + dQ_3^+}{dV} + \frac{dQ_2^+}{dV}$$

$$= \frac{|dQ_3^-|}{|dV|} + \frac{|dQ_3^+|}{|dV|} + C_j(V)$$

$$Q_3^+ = Q_{P_\infty} = qA\int_{x_n}^{\infty} \Delta p_n(x)\,dx$$

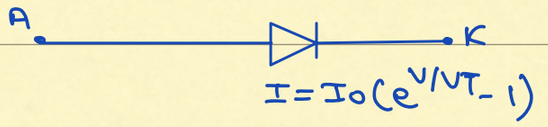
Similarly $Q_3^- = qA L_n n_{p0}(e^{V/V_T}-1)$

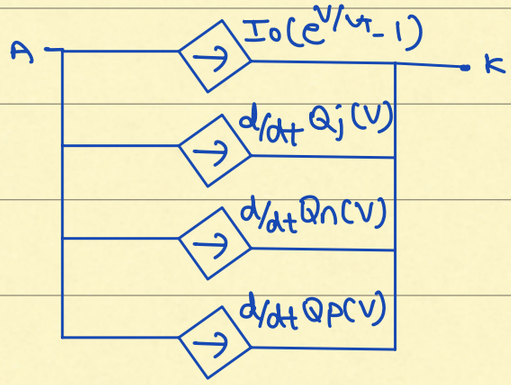
$$\Delta p_n(x) = \Delta p_n(x_n) e^{-(x-x_n)/L_p}$$

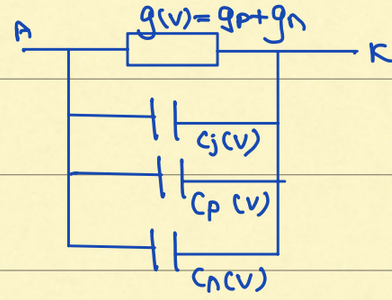
$$\frac{Q_p}{\tau_p} + \frac{Q_n}{\tau_n} = qA\left(\frac{D_p}{L_p}p_{n0} + \frac{D_n}{L_n}n_{p0}\right)(e^{V/V_T}-1)$$

$$C_{diode} = \frac{|dQ_3^-|}{dV} + \frac{|dQ_3^+|}{dV} = \frac{dQ_n}{dV} + \frac{dQ_p}{dV} = g_n\tau_n + g_p\tau_p \approx g\tau$$

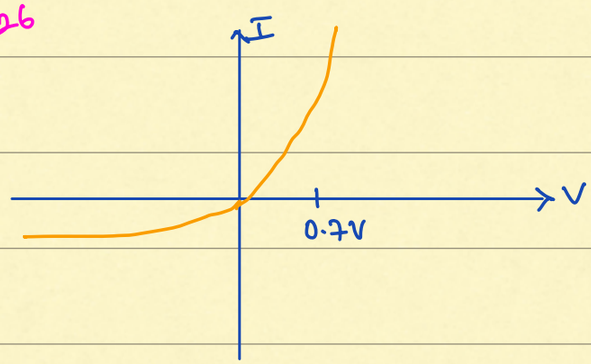
g → conductance

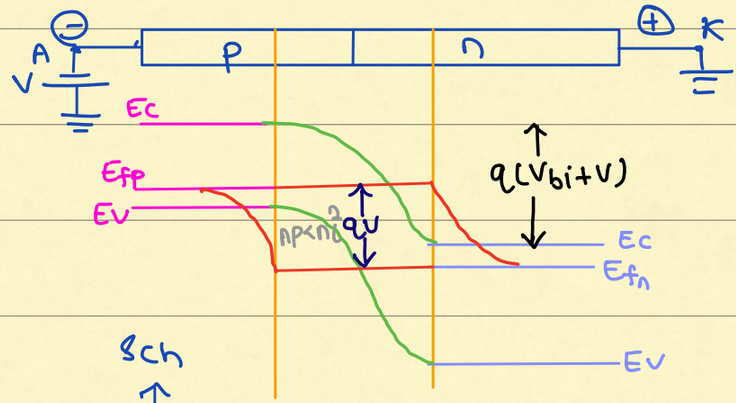


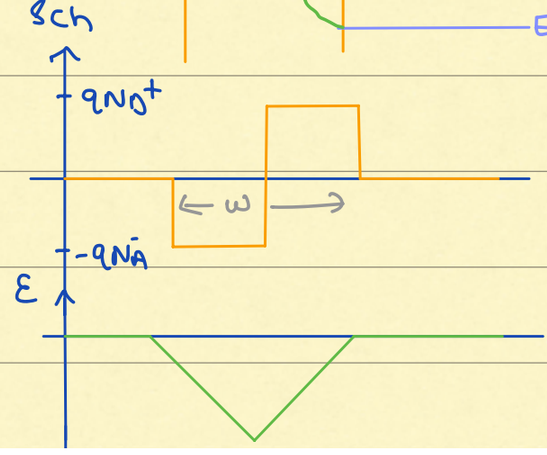




3/27/26

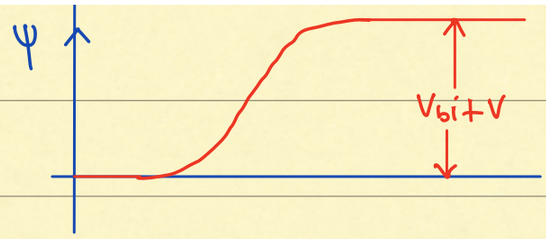






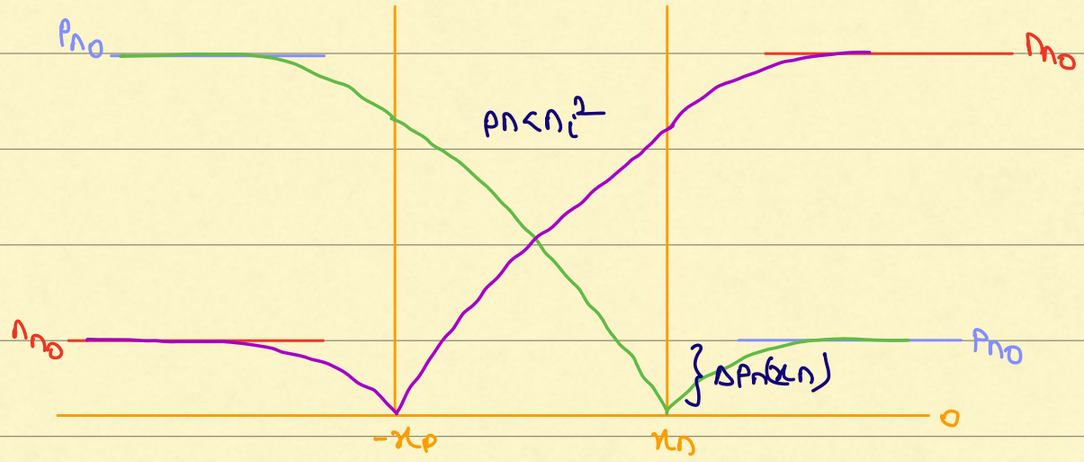
$$W = \sqrt{\frac{2\epsilon_{si}(V_{bi}+V)}{qN_{eff}}}$$

$$\mathcal{E}_{peak} = \frac{2(V_{bi}+V)}{W}$$

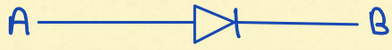


$$\Delta p_n(x) = \Delta p_n(x_n) e^{-(x-x_n)/L_p}$$

$$\Delta p_n(x_n) = p_{n0}(e^{V/V_T} - 1)$$



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$$W = \sqrt{\frac{2\epsilon_{Si}(V_{bi}-V)}{qN_{eff}}}$$

Assumptions:

1. All applied potential is dropped across SCR
2. Low level injection
3. No net generation / recombination in SCR

$$\Delta p_n(x_n) = p_{n0}(e^{V/V_T} - 1)$$

$|J_{p,drift}| \approx |J_{p,diffusion}|$ in SCR

$|J_{n,drift}| \approx |J_{n,diffusion}|$ in SCR

$$J_n(-x_p) = J_n(x_n)$$

$$J_p(-x_p) = J_p(x_n)$$

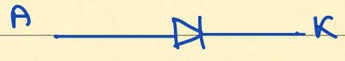
$$J_n(x) = J_n(-x_p) + J_n(x_n)$$

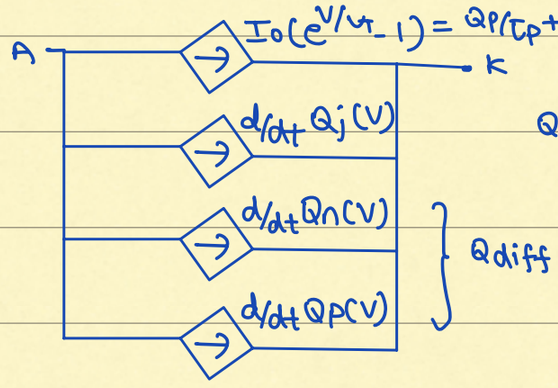
$$p_n(x_{p0}) = p_p(-x_{p0})\, e^{-V_{bi}/V_T}$$

$$p_{n_0} = p_{p0} \exp\left(\frac{-V_{bi}}{V_T}\right) \qquad V_{bi} = \frac{KT}{q} \ln\left(\frac{p_{p0} n_{n0}}{n_i^2}\right)$$

$$p_n(x_n) = p_p(-x_p) \exp\left(-\frac{V_{bi} - V}{V_T}\right)$$

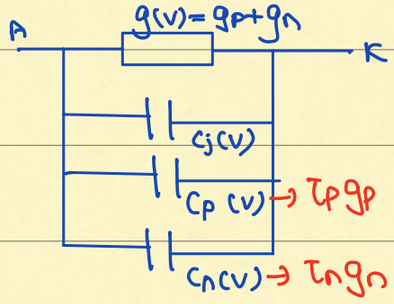
p_{p0}





$$Q_j = \int_0^V C_j(V)\, dV$$

$$C_j(V) = \frac{C_{j0}}{\sqrt{1 - V/V_{bi}}} = \frac{\varepsilon A}{W}$$



$$C_{diff}(V) = \frac{\tau_p I_p}{V_T} + \frac{\tau_n I_n}{V_T}$$

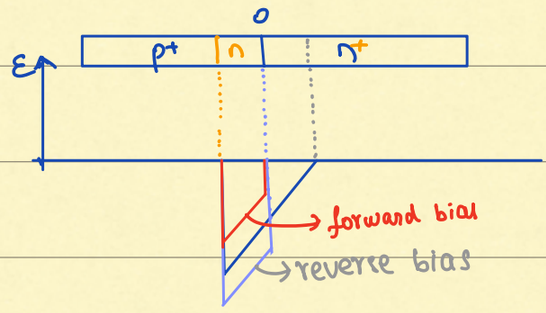
$p^+ n n^+$ $p^+ p n^+$

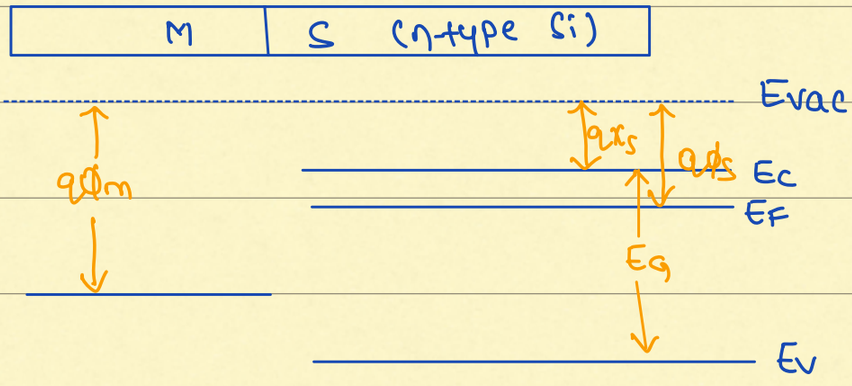
$\Downarrow$

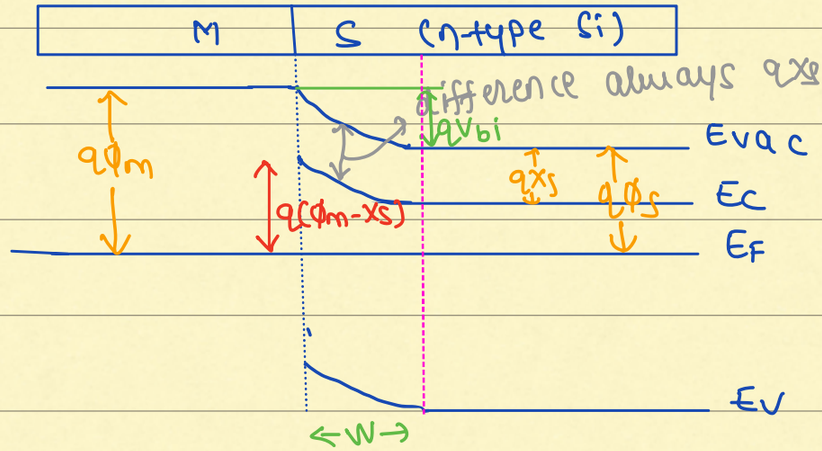
$p^+ n$ diode

$N_A = 10^{18}/cc$ $N_A >>> N_D$

$N_D = 10^{15}/cc$







qχs can't change (material property)

$$W = \sqrt{\frac{2\varepsilon_{Si} V_{bi}}{qN_D}}$$

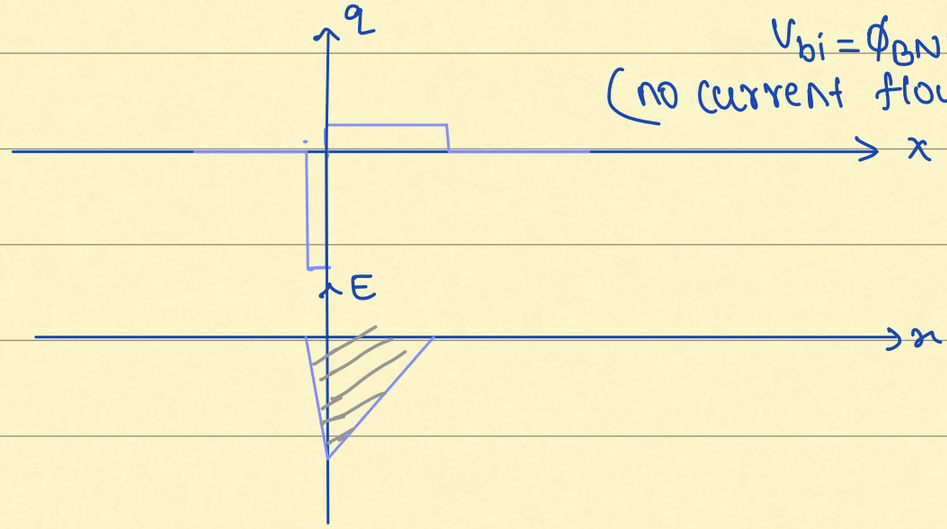
$$V_{bi} = \phi_m - \phi_s$$

$$= \phi_{BN} - (\phi_s - \chi_s)$$

$$\phi_s = \frac{E_c - E_v}{q}$$

barrier of metal to n-type semiconductor $q\phi_{Bn}$

$$q\phi_{Bn} = q(\phi_m - \chi_s)$$



$$V_{bi} = \phi_{BN} - V_T \ln\left(\frac{N_C}{N_D}\right)$$

(no current flows from M to S)

At surface of semiconductor $n_s = N_D \exp\left(\frac{-V_{bi}}{V_T}\right)$

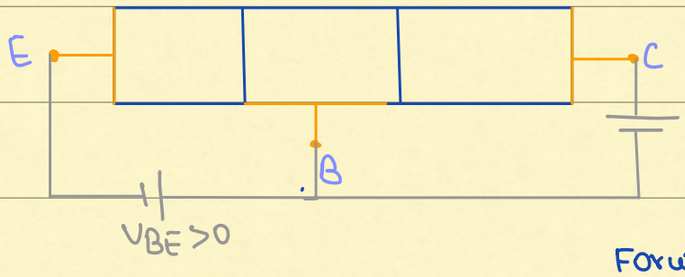
$n_s = N_c \exp\left(\frac{-\phi_{Bn}}{V_T}\right)$

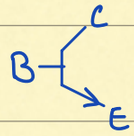
At equilibrium $\Rightarrow$ $|J_{m-s}| = |J_{s-m}| = C_1 n_s$ (coulomb × velocity)

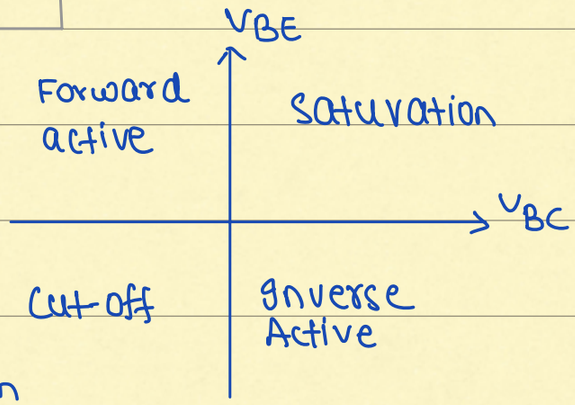
forward bias: $J = J_{s-m} - J_{m-s}$

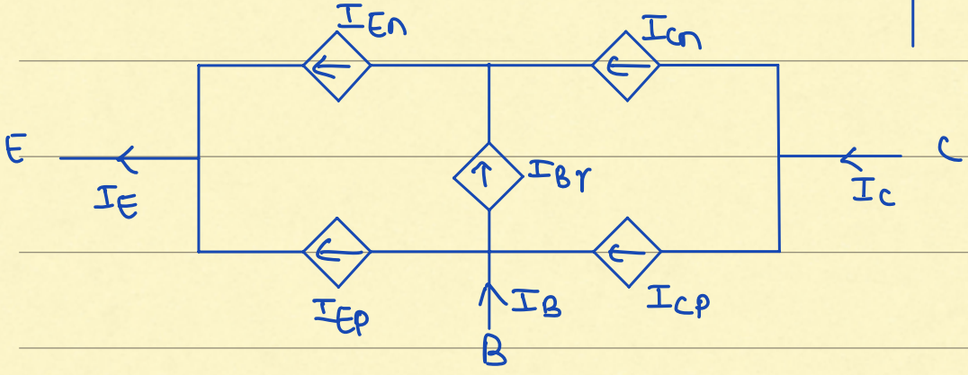
↳ barrier height reduces so $e^{\ominus}$ flow becomes easier

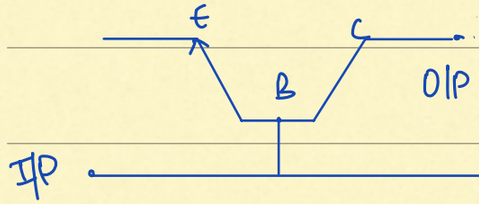
$= C_1 N_c \exp\left(\frac{-\phi_{Bn}}{V_T}\right)\left[\exp\left(\frac{V}{V_T}\right) - 1\right]$





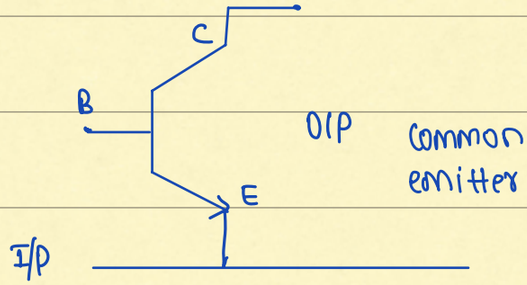


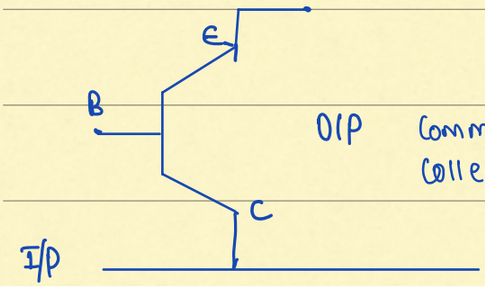




$V_{BC} = 0$

$I_{Cp} = 0$





$$\alpha = \frac{I_C}{I_E}$$

common base current gain

$$\beta = \frac{I_C}{I_B}$$

common emitter current gain

$$\gamma = \frac{I_{En}}{I_E}$$

emitter injection efficiency

$$\alpha_T = \frac{I_{Cn}}{I_{En}}$$

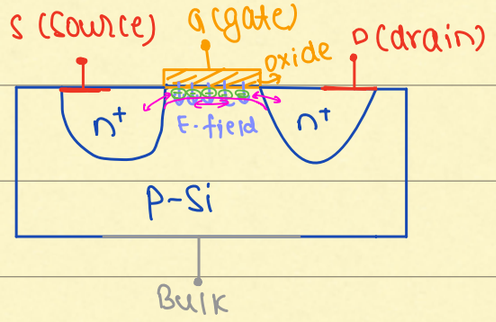
base transport factor

$$I_{EP} = \frac{qAD_{PE}\,\Delta p_{nE}}{W_E} = qAD_{PE}\frac{n_i^2}{N_{DE}W_E}\left(e^{V_{BE}/V_T}-1\right)$$

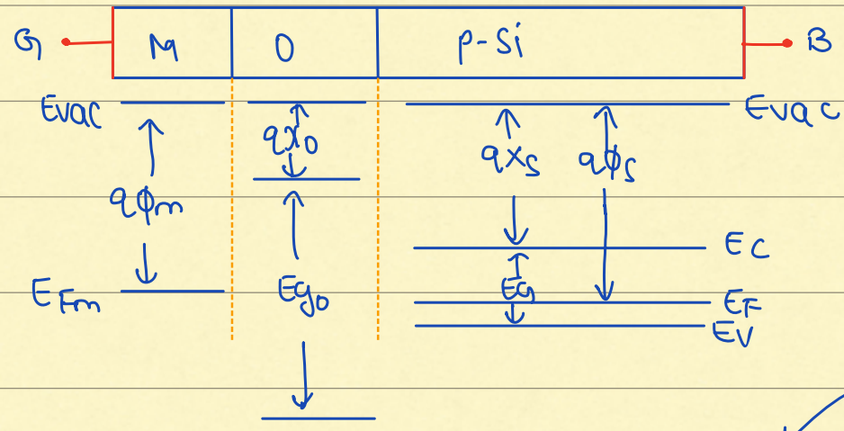
$$I_{En} = qAD_{nB}\frac{\Delta n_{pB}}{W_B} = qAD_{nB}\frac{n_i^2}{N_{AB}W_B}\left(e^{V_{BE}/V_T}-1\right)$$

$$\gamma = \frac{1}{1+\left(\frac{D_{PE}}{D_{nB}}\right)\left(\frac{N_{AB}}{N_{DE}}\right)\left(\frac{W_B}{W_E}\right)}$$

MOSFET metal oxide semiconductor field effect transistor



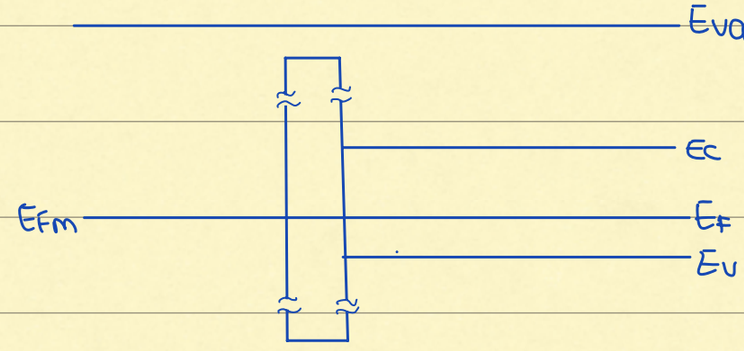
if bias is put b/w G & B
G (+ve) & B (GND)



as they are joined (assumptions: $\phi_m = \phi_s$; $Q_{oxide} = 0$)

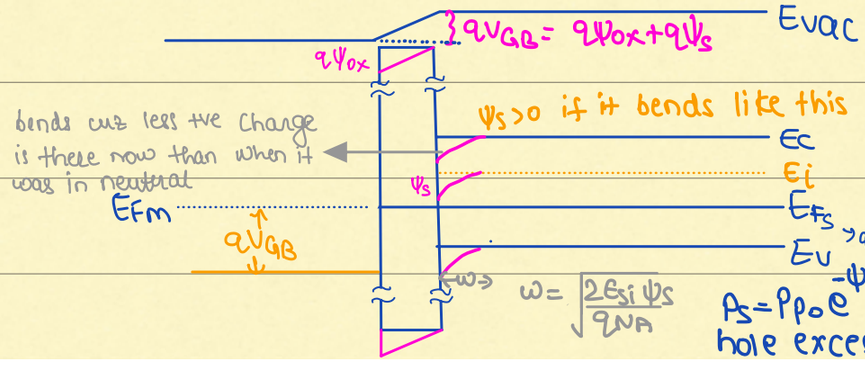
Ideal MOS

$Q_{oxide} = 0$ → charge accumulated in oxide



Flat band situation

Now if $V_G > 0$ and $V_B = 0$



Interface b/w SiO_2 & Si

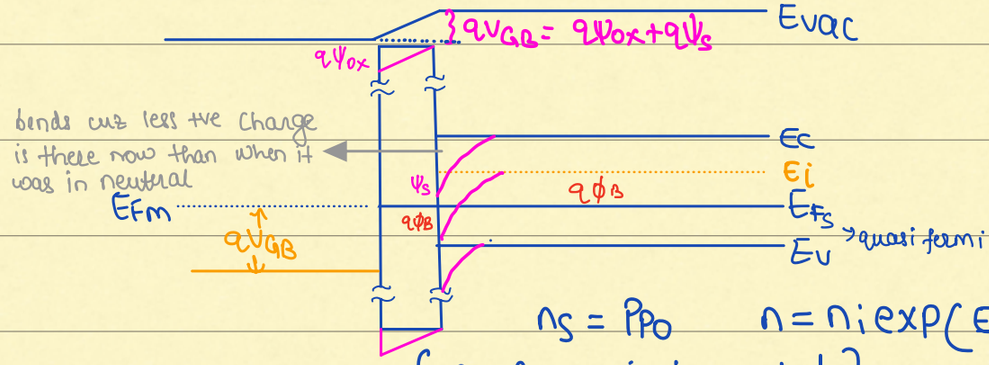
ψ_s → surface potential

E_{Fs} → quasi fermi

$p_s = p_{p0} e^{-\psi_s / V_T}$

hole excess @ surface

$$N_A = n_i \exp\left(\frac{E_i - E_{FS}}{KT}\right) \Rightarrow \phi_B = V_T \ln\left(\frac{N_A}{n_i}\right)$$



$n_s = p_{po}$ $\quad n = n_i \exp\left(\frac{E_{fn} - E_i}{KT}\right)$

{ surface is inverted strongly }



$$p_s = p_{po} \exp\left(\frac{-\psi_s}{V_T}\right)$$

$$n = n_{po} \exp\left(\frac{\psi_s}{V_T}\right)$$

$Q_{semic} = Q_{bulk} + Q_{electron}$ — just a layer

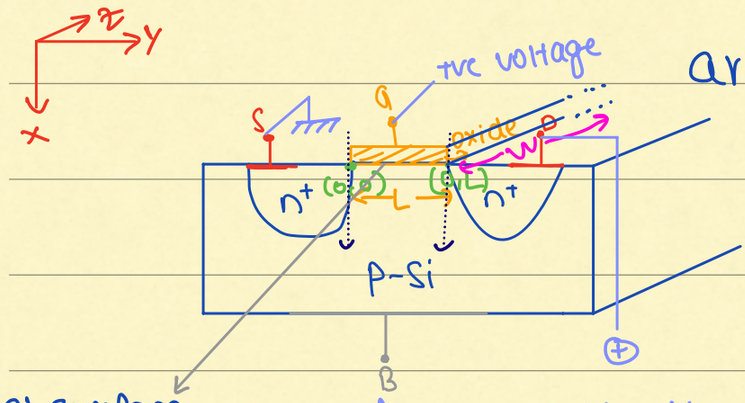
$\approx Q_{bulk} = -qN_AW$

$$Q_s = Q_B = -\sqrt{2qN_A\varepsilon_{Si}(2\phi_B)}$$

$Q_B \propto \sqrt{\psi_s}$

$Q_n \propto e^{(\psi_s/V_T)}$

V_{GB} for which strong inversion happens is called MOS threshold voltage



area of Oxide layer = LW

L/W ratio is less

{ 3nm processor in fone
⇒ L = 3nm }

at surface

$n_s = N_A = p_{p_0}$

If $V_B > 0 \Rightarrow V_{BS}, V_{BD}$ both are fwd bias so current starts flowing

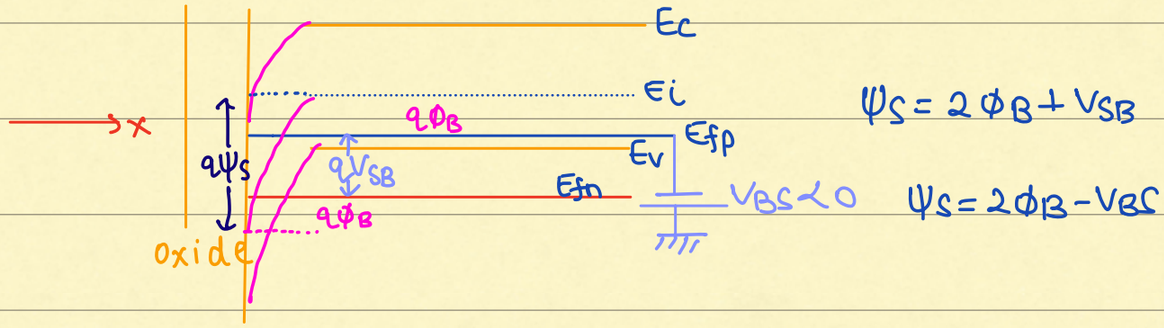
So we take $V_{BS} < 0$

we draw energy band diagram at (0,0) and (0,L)

$$n = n_i \exp\left(\frac{E_{fn} - E_i}{KT}\right) \quad p = n_i \exp\left(\frac{E_i - E_{fp}}{KT}\right)$$

$$np = n_i^2 \exp\left(\frac{E_{fn} - E_{fp}}{KT}\right)$$

@ y=0 line



$\psi_s = 2\phi_B + V_{SB}$

$\psi_s = 2\phi_B - V_{BS}$

@ y=L line $\quad \psi_s = 2\phi_B + V_{DB} = 2\phi_B + V_{DS} - V_{BS}$

@ any cutline by y=0 & y=L : $\psi_s = 2\phi_B + V_{CB}(y)$

$= 2\phi_B + V_{CS}(y) - V_{BS}$

Threshold voltage for MOSFET is defined @ $V_{DS} = 0$

$$V_{GB} - V_{FB} = \psi_s + \psi_{ox} = \psi_s + \left(-\frac{Q_s}{C_{ox}}\right)$$

→ Till threshold condition (strong inversion)

$n_S \leq p_{P0}$ $\quad Q_S = Q_n + Q_B \approx Q_B$

$$\Rightarrow V_{GS} - V_{BS} - V_{FB} = \psi_S + \frac{\sqrt{2q N_A \epsilon_{Si} \psi_S}}{C_{ox}}$$

→ If $V_{GS} = V_{TH}$; $n_S = p_{P0}$; $\psi_S = 2\phi_B - V_{BS}$

$$\Rightarrow V_{TH} = V_{FB} + 2\phi_B - V_{BS} + \frac{\sqrt{2q N_A \epsilon_{Si} (2\phi_B - V_{BS})}}{C_{ox}}$$

$$\Rightarrow V_{TH} = V_{FB} + 2\phi_B + \gamma\sqrt{2\phi_B - V_{BS}}$$ body factor γ

for capacitor : $V_{TH} = V_{FB} + 2\phi_B + \gamma\sqrt{2\phi_B}$

$$J_n(x,y) = n(x,y)\mu_n \frac{d}{dy}(E_{Fn}(y) - E_{fp})$$ → anyways always zero

$$= -qn(x,y)\mu_n \frac{d}{dy}[V_{CB}(y)]$$

$$I_n(y) = W\int_0^{x_{inv}} J_n(x,y)\,dx = W\int_0^{x_{inv}} -qn(x,y)\mu_n \frac{d}{dy}[V_{CB}(y)]\,dx$$

$$= W\mu_n \frac{d}{dy}[V_{CB}(y)] \int_0^{x_{inv}} -qn(x,y)\,dx$$

$$= W\mu_n \frac{d}{dy} V_{CB}(y) \cdot Q_n(y)$$ inversion charge per unit area

$$\int_0^L I_n(y)\,dy = -L I_{DS} = \mu_n W \int_0^L \frac{d}{dy} V_{CB}(y)\, Q_n(y)\,dy$$

I_{DS} ↓ drain to source current

$$= \mu_n W \int_{-V_{BS}}^{V_{DS}-V_{BS}} Q_n(V_{CB})\,dV_{CB}$$

$$\Rightarrow I_{DS} = -\frac{\mu_n W}{L} \int_{-V_{BS}}^{V_{DS}-V_{BS}} Q_n(V_{CB})\,dV_{CB}$$

$Q_S = Q_n + Q_B$ $\qquad Q_B = -\sqrt{2qN_a\epsilon_{si}(2\phi_B + V_{CB})}$

$$Q_n(V_{CB}) = Q_S - Q_B = -C_{ox}[V_{GB} - V_{FB} - 2\phi_B - V_{CB}] + \sqrt{2qN_A\epsilon_{si}(2\phi_B + V_{CB})}$$

put $Q_n(V_{CB})$ back and integrate we get

level II SPICE MOSFET